

40.520 Stochastic Modeling

Feb 25, 2026

Mid-term examination

Full Marks - 55

- Attempt all questions. Questions may appear on either side of a sheet.
- The exam is 120 minutes in duration. The maximum score is 55 points.
- The use of a non-programmable scientific calculator and an A4 sheet with handwritten information on one side is allowed.
- Write your answers in this booklet. Request for extra paper only when needed.
- Return the extra sheets also when returning the exam
- Sign the SUTD honour code in the back of this page.

Question 1. ($5 \times 2 = 10$ points) Let $(X_n : n \geq 0)$ be an irreducible, discrete time Markov chain with countably infinite state space I , probability transition matrix $P = ((p_{ij})), i, j \in I$. Considering this Markov Chain, circle around the right answer (TRUE or FALSE) for questions below. Ambiguous answers and unclear markings will result in zero points. You do NOT have to justify your answer for the objective questions in Parts (a) - (e) of Question Q1.

- (a) It is possible for all states to be transient. **TRUE/ FALSE**
(b) If the equation $\pi = \pi P$ has no solution then all states are transient. **TRUE/ FALSE**
(c) If $p_{ii} = 0$ for all $i \in S$, then the chain must necessarily be periodic. **TRUE/ FALSE**
(d) If $p_{ii} > 0$ for all $i \in S$, then a stationary distribution must exist for P . **TRUE/ FALSE**
(e) An invariant distribution exists when the state space is $I := \{0, 1, 2, \dots\}$ and transition probabilities are given by,

$$p_{i,j} = \begin{cases} \frac{1}{2} & \text{if } i \geq 2 \text{ and } j \in \{i-1, i+1\}, \\ 1 & \text{if } i+j = 1. \end{cases}$$

TRUE/ FALSE

Question 2. ($7 + 7 + 1 = 15$ points) Smith is in jail and has 3 dollars; he can get out on bail if he has 8 dollars. A guard agrees to make a series of bets with him. If Smith bets A dollars, he wins A dollars with probability 0.4 and loses A dollars with probability 0.6. Find the probability that he wins 8 dollars before losing all of his money if,

- a) he bets 1 dollar each time (timid strategy);

- b) he bets, each time, as much as possible but not more than necessary to bring his fortune up to 8 dollars (bold strategy);
- c) which strategy gives Smith the better chance of getting out of jail?

Question 3. (3 + 5 = 8 points) Suppose the vertices of a triangle are labelled 1, 2 and 3, and Tom (a cat) and Jerry (a mouse) move around these vertices in the following way. If Tom is at vertex i and Jerry is at vertex j , where $i \neq j$, then Tom will move to vertex j with probability $p \in (0, 1)$ and to the other vertex, say k , with probability $1 - p$. If Jerry sees Tom moving to vertex j , then Jerry instantaneously jumps to vertex k ; otherwise, Jerry remains at vertex j (so Tom never catches Jerry).

- a) Let X_n denote the location of Jerry after n movements of Tom. Does $(X_n : n \geq 0)$ form a Markov chain?
- b) Determine the probabilities of finding Jerry in Vertices 1, 2, and 3.

Question 4 (4 + 2 = 6 points) A saleswoman flies around between Singapore, Bangkok, and Jakarta as follows. From Singapore she goes to Bangkok or Jakarta with probability $1/2$ each; from Bangkok she goes to Singapore with probability $3/4$ and to Jakarta with probability $1/4$; from Jakarta she always goes back to Singapore.

- a) What is the probability of finding the saleswoman in Singapore?
- b) What fraction of her trips are from Bangkok to Singapore?

Question 5. (8 points) Suppose that $0 < p < q := 1 - p < 1$ for the discrete-time Markov chain model on the state-space $\{0, 1, 2, 3, \dots\}$ with transition probabilities given by $p_{0,1} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $i = 1, 2, \dots$. Find the expected time it takes for the chain to reach state '0' when the chain is began at state '1' at time 0.¹

Question 6. (8 points) Let $(X_n : n \geq 0)$ be Markov(λ, P). Suppose that we can observe the chain X_n only when it moves. So the observable process $(Z_m : m \geq 0)$ is given by

$$Z_m = X_{S_m},$$

where $S_0 = 0$ and

$$S_{m+1} = \min\{n \geq S_m : X_n \neq X_{S_m}\}.$$

Is $(Z_m : m \geq 0)$ a Markov chain? Give reasons for your answer. Identify the transition probabilities if you argue it to be a Markov chain.

¹Hint: **Question 5** – Let k_i be the expected time it takes to reach state 0 when the Markov chain is initiated at state i . If you try to do first-step analysis, you will get an infinite system of linear equations. To solve the equations, you may use the symmetry in the problem to guess how k_i should behave as a function of i ? In particular, how many times you have to add k_1 to get k_2 ? How many times you have to add k_1 to get k_3 ?, and so on.